

Problem Set 3 - Variables and Data types

Hello student!

The goal of this problem-set is to familiarize with the different data-type.

As you learned in the lectures, despite it is not required to specify the data type when declaring a variable, each variable has its data-type.

The available data-types are:

- Numeric: `int`, `float`, `complex`
- Text: `string`
- Sequence: `list`, `tuple`, `range`
- Set: `set`, `frozenset`
- Mapping: `dictionary`
- Boolean: `bool`
- Binary: `bytes`, `bytearray`, `memoryview`

If you want to learn more about the different types, check the ["Built-in Types" documentation](#).

Remember, you can check the data type of a variable using the function `type()`:

```
my_var = 14.2
print(type(my_var))

<class 'float'>
```

Assignment 3.1 - Divisible (optional)

We want to develop a program that is able to tell us if a number is a divisor of another number. The two numbers must be requested to the user. The program must then print if the first inserted is a divisor of the second.

The output must be the following (in red you see the values inserted by the user):

```
Please insert a number: 6
Please insert another number: 18
Is the first number a divisor of the second? True
```

Assignment 3.2 - Calculator (optional)

Write a program that asks the user for two decimal values (e.g. 3.1415), and then calculates and prints the following (in red the values inserted by the user):

```
First value: 3.2
Second value: 10.6
Quotient: 3.2 / 10.6 = 0.30188679245283023
Product: 3.2 * 10.6 = 33.92
Sum: 3.2 + 10.6 = 13.8
Difference: 3.2 - 10.6 = -7.3999999999999995
```

Assignment 3.3 - String manipulation (optional)

Write a program that asks the user for a string and prints it with every word capitalized and without leading and trailing whitespaces. The wrongly inserted `**` chars must also be substituted.

For example, if the user inserts the following:

```
my**NaMe**is**WaLter**haRTwell**WHite.
```

The output must be:

```
My Name Is Walter Hartwell White.
```

Hint: check the ["String Methods" documentation](#) for all the methods available on strings.

Assignment 3.4 - Value swap (optional)

Develop a software asking for two floats. The values must be printed and then they must be swapped between variables. The final value must then be printed.

Example of program execution with the output (in red the input inserted by the user):

```
Insert first value: 1.234
Insert second value: 6.789
Values before swapping:
    first_val: 1.234
    second_val: 6.789
Values after swap:
    first_val: 6.789
    second_val: 1.234
```

Important: the values have to actually be swapped between the first and the second variable!

Assignment 3.5 - Circle (optional)

Develop a software asking for the radius of a circle as a float number, then calculate and print the values of the diameter, circumference, and the area of the circle.

Remember:

- Diameter: $d = 2 \times r$
- Circumference: $C = 2 \times \pi \times r$
- Area: $A = \pi \times r^2$

Example of program execution with the output (in red the input inserted by the user):

```
Insert circle's radius: 42
Diameter: 84.0
Circumference: 263.89378290154264
Area: 5541.769440932395
```

Hint: almost every programming language provides a math package that contains the value of the constant π . To import it in Python use:

```
from math import pi
```